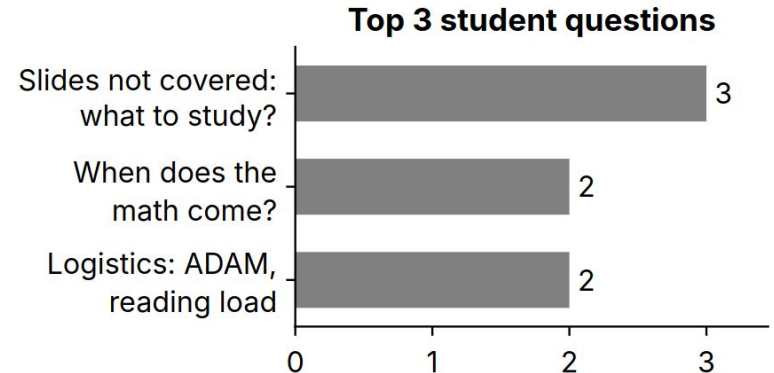
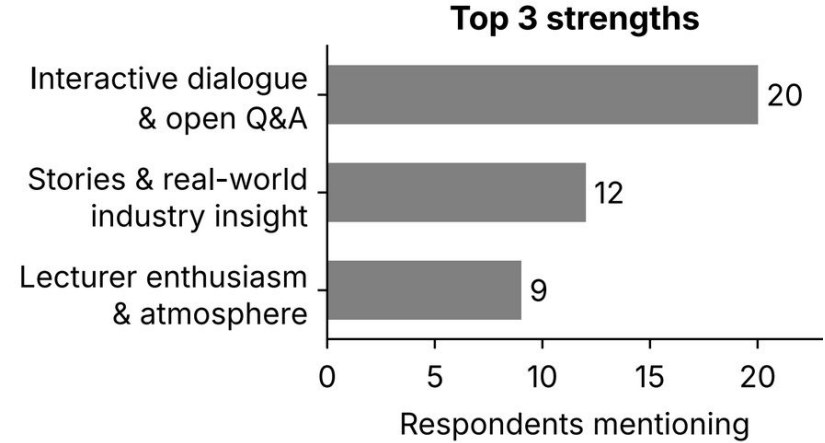
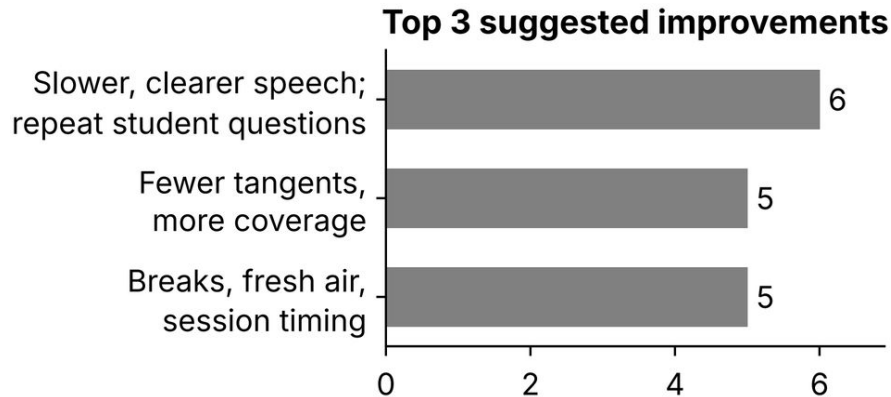
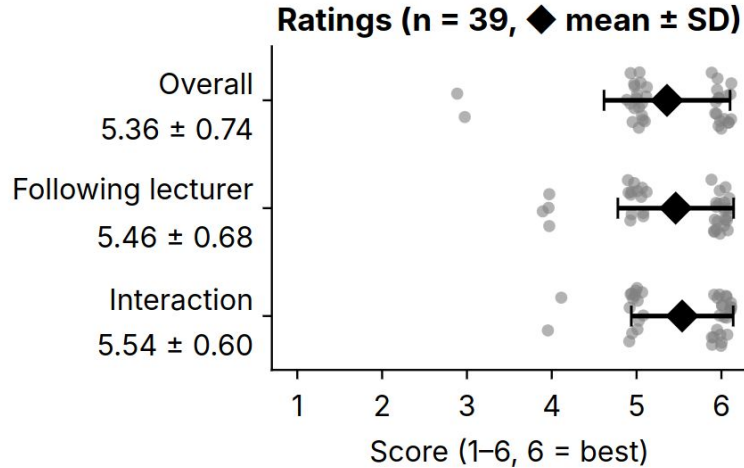
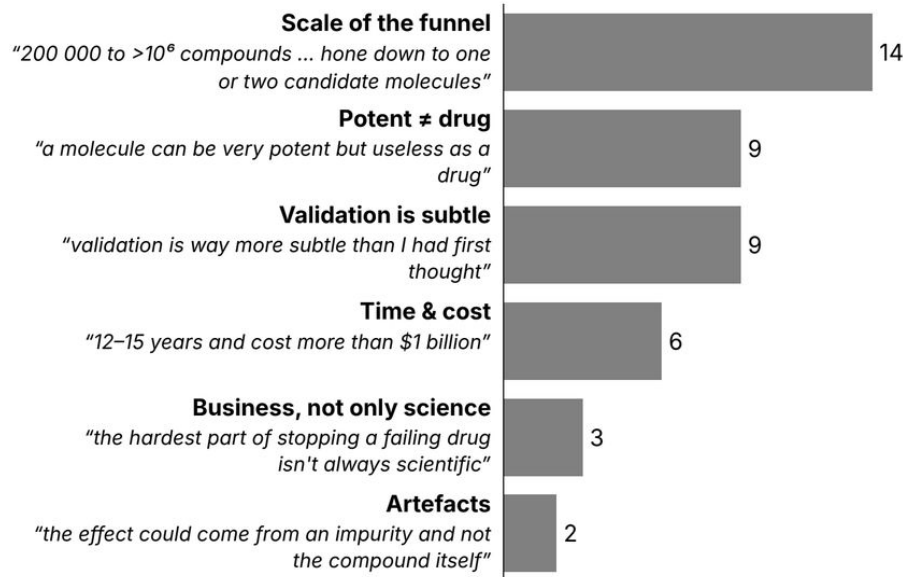


Feedback to lecture 2

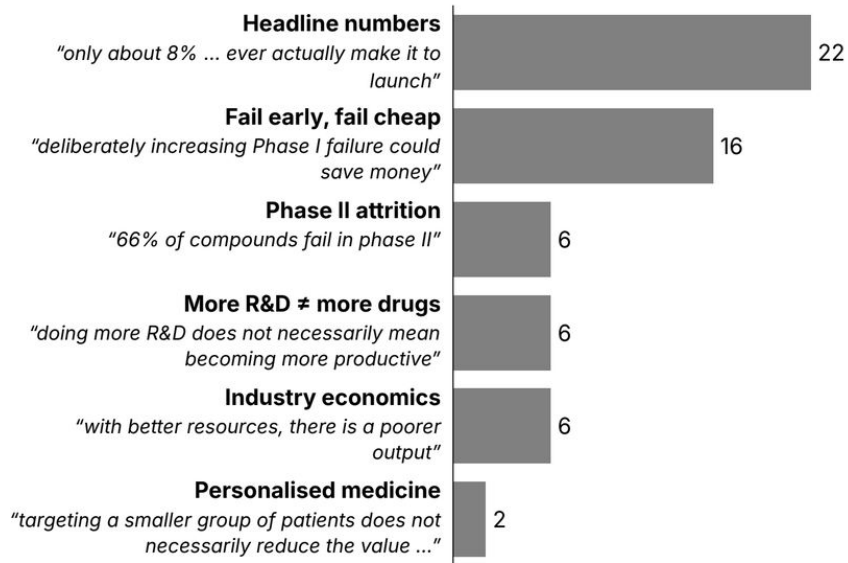


Summary of the most learnings

Hughes et al. (2010): what surprised you most?

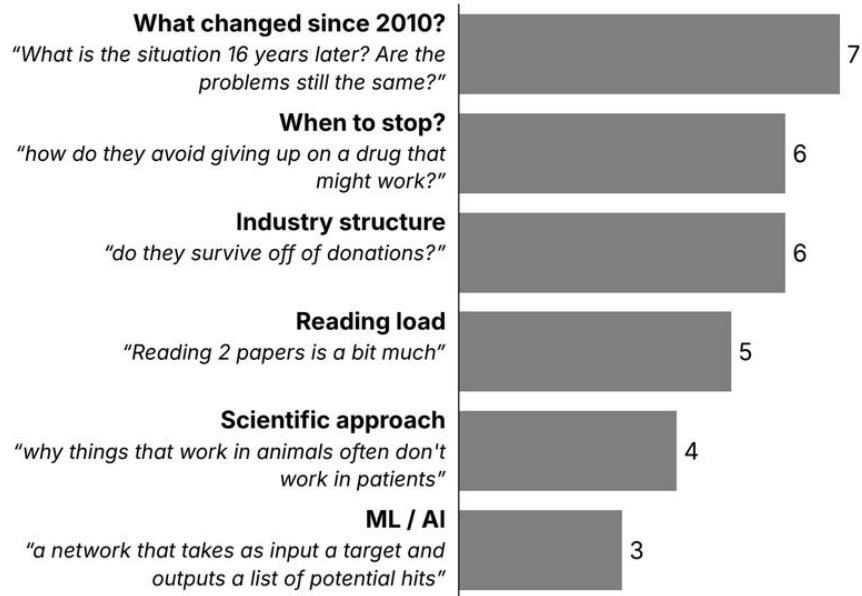


Paul et al. (2010): what surprised you most?



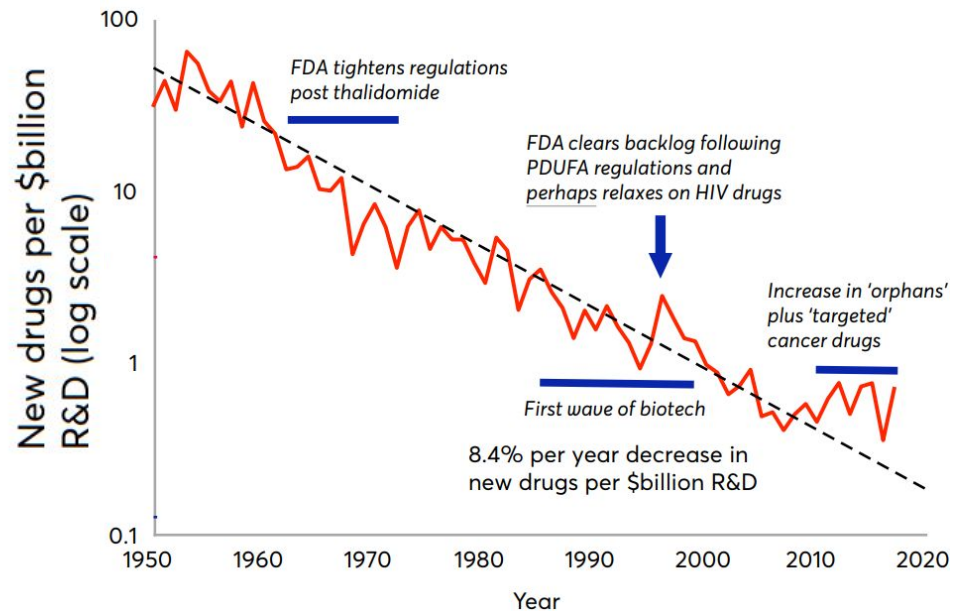
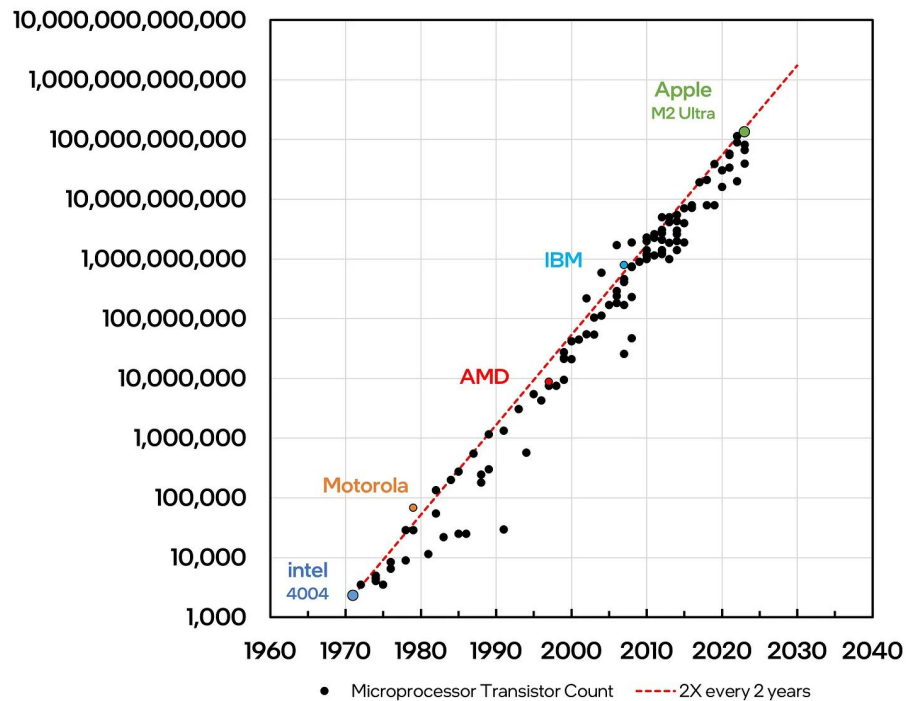
Selected questions and comments

- If we consider drug discovery as an optimization problem, what it is trying to optimize? (David's framing) Are there potential issues associated with the optimization view? Will pharma [learn the bitter lesson](#): computational power outperforms domain understanding in the long run? - I don't know, have my bet though.
- Why things work in animals often do not work in human? - There are many reasons. To start, disease models are not diseases.
- **When and how to stop? How about false negatives?** I wish I knew the answer.
- How about failed projects: are there efforts sharing the molecules and learnings? - Yes, for instance [Donated Chemical Probes](#).
- Views about generics and country-specific regulatory interactions are ignored - YES.



How to read faster? - Ask, Read, Explain, Repeat.
Until then: In case you find it is too much reading and you did not read or finish, please reply so - you are not judged.

Moore's versus Eroom's Law

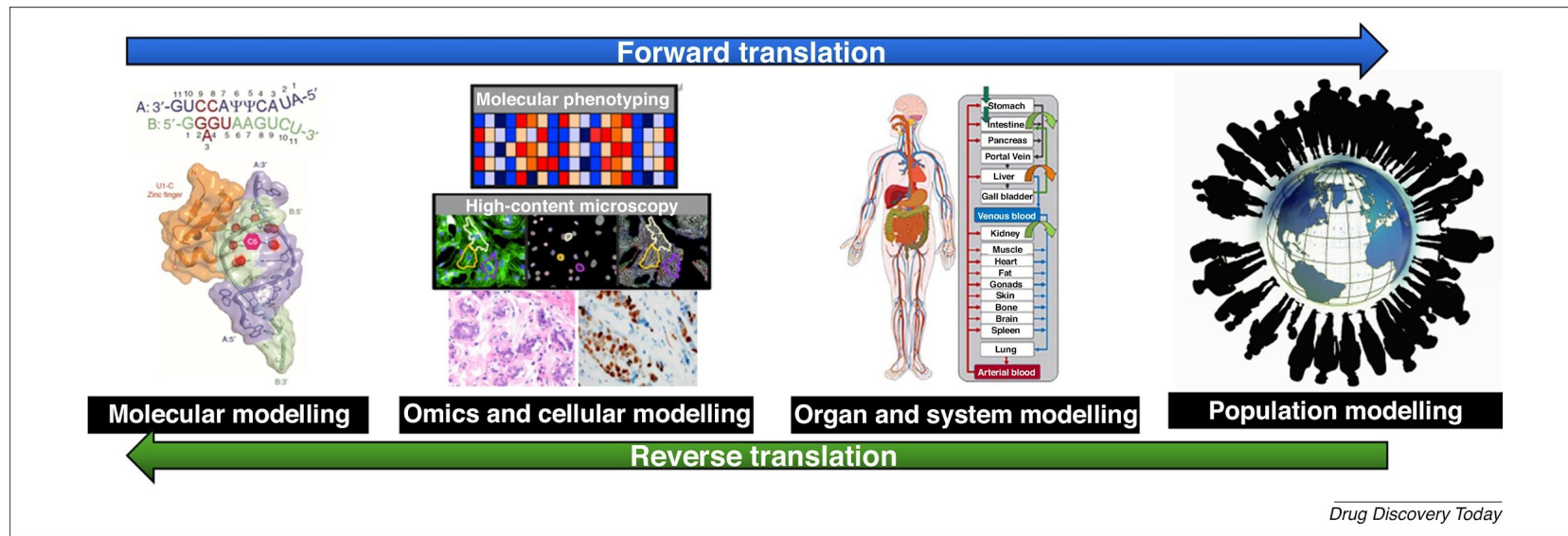


Data come from Scannell, et al. (2012) Diagnosing the decline in pharmaceutical R&D efficiency. Nature Reviews Drug Discovery, and personal communication. Figure by [Richard Jones and James Wilsdon](#)

Adapted from graphs by [Hannah Ritchie and Max Roser](#) and by [Jim Keller](#)

AMIDD 2026 Lecture 3:

The *What, Who, and How* (continued)



PD Dr. Jitao David Zhang, Computational Biologist

¹ Computational Sciences Center of Excellence (CS CoE), Roche Innovation Center Basel, F. Hoffmann-La Roche;

² Department of Mathematics and Computer Sciences, University of Basel

Outline

1. *How*: The multi-scale views of drug discovery
2. *What*: Modalities
3. *Who*: A role game

Glucagon-like peptide 1 (GLP-1) and chemical derivatives

GLP-1 is a 30- or 31-amino-acid peptide. It is derived from a protein (preproglucagon) encoded by the gene GCG in the human genome.

Endogenous GLP-1 has a half-life of about 2 minutes, making the natural form a poor drug candidate due to its pharmacokinetic (PK) behaviour.

Synthesized agonists have similar sequences but much longer half-lives due to chemical modifications.

Examples include:

- Semaglutide (top, Novo Nordisk, approved, half-life: about 1 week)
- Taspoglutide (bottom panel, tested in Phase III trials, discontinuation due to hypersensitivity and poor GI tolerability)

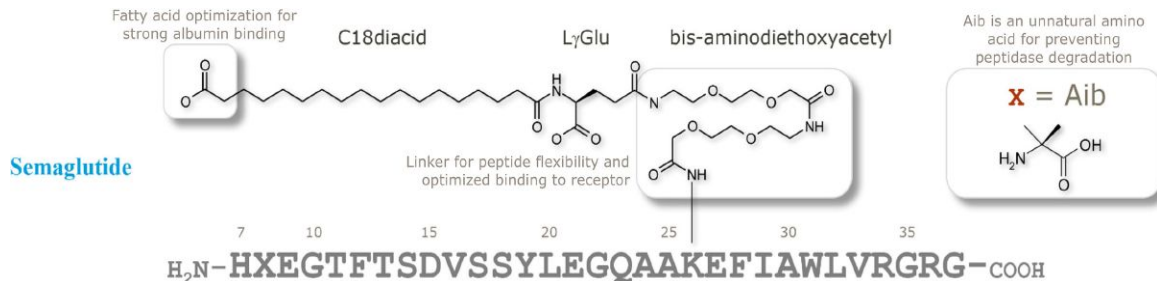


Figure from [Knudsen et al., 2019](#)

SEMAGLUTIDE	1 HXEGTFTSDVSSYLEGQAAKEFIAWLVRGRG	31
GLP-1	1 HAEGTFTSDVSSYLEGQAAKEFIAWLVRGRG	31
TASPOGLUTIDE	1 HXEGTFTSDVSSYLEGQAAKEFIAWLVRGRG	30
GLP-1	1 HAEGTFTSDVSSYLEGQAAKEFIAWLVRGR	30

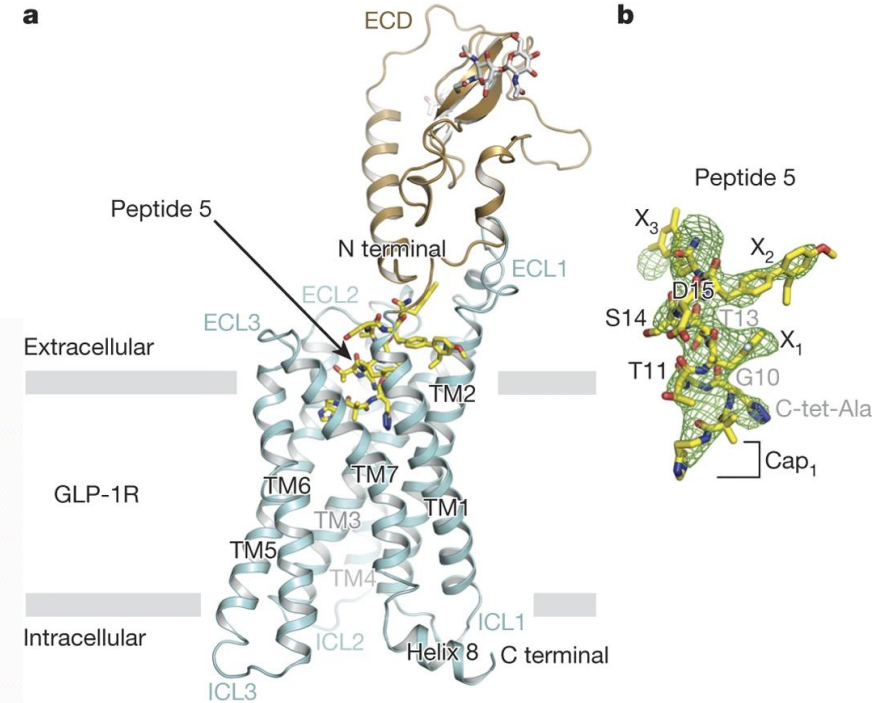
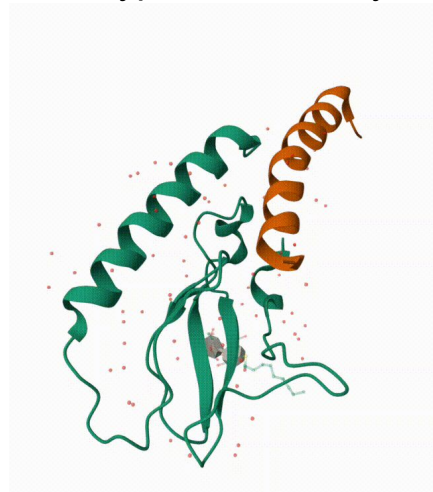
GLP-1 binds to its receptor GLP1R to induce insulin secretion

GLP-1 is secreted from the intestine after nutrient intake. GLP-1 binds to its receptor GLP1R, and the binding event activates intracellular signaling and stimulates glucose-dependent insulin secretion.

GLP1R is a GPCR (G protein-coupled receptor) found on pancreatic β cells and other cell types in the body.

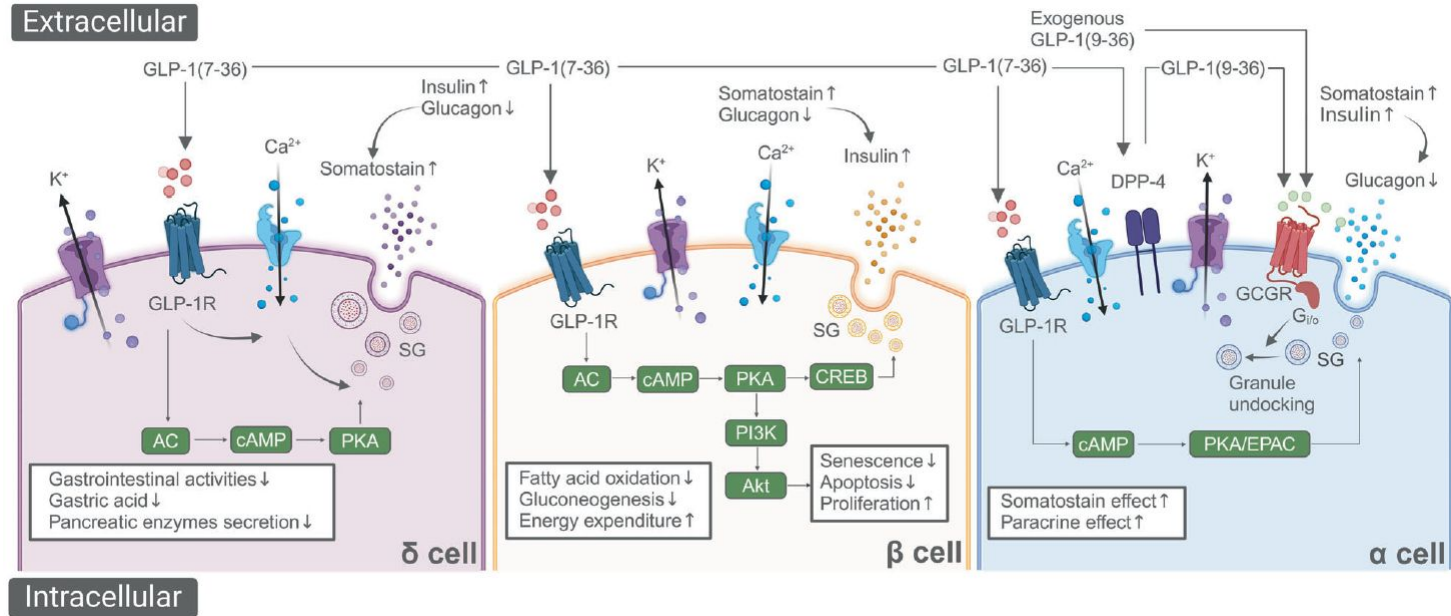
Left: crystal structure of GLP-1 (orange) bound to a fragment of GLP1R (green) (PDB: [3IOL](#))

Right: ribbon representation of GLP1R bound to an agonist peptide. The transmembrane domain, or TMD, is coloured cyan. The extracellular domain, or ECD, is coloured brown.



A. Jazayeri et al. (2017), "Crystal structure of the GLP-1 receptor bound to a peptide agonist," <https://doi.org/10.1038/nature22800>, Nature

Mechanisms of blood glucose reduction by GLP-1 in pancreatic α (alpha), β (beta), and δ (delta) cells



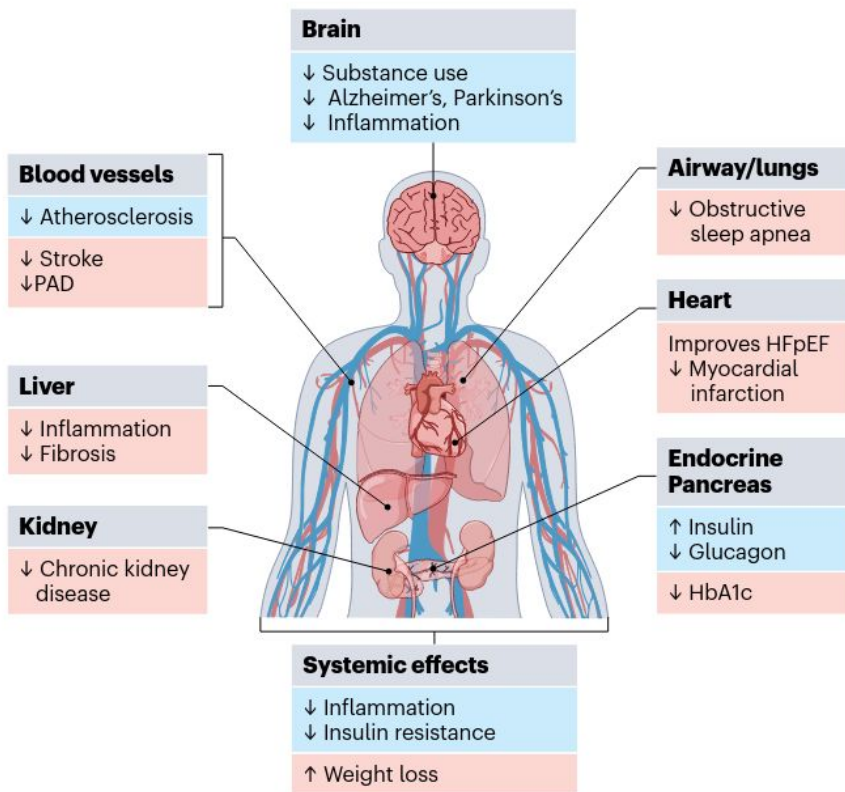
The cartoons give a simplified summary of our current understanding of how GLP-1 reduces blood glucose levels by acting on different pancreatic cell types. Direct and indirect effects of GLP-1 on other cells may employ other mechanisms.

Organ-level effects of GLP-1 medicines: established and potential benefits

The benefits of GLP-1 medicines have been demonstrated in multiple outcome trials for people with

- cardiovascular disease,
- peripheral artery disease (PAD),
- liver disease, kidney disease,
- Heart Failure with preserved Ejection Fraction (HFpEF), and
- obstructive sleep apnea

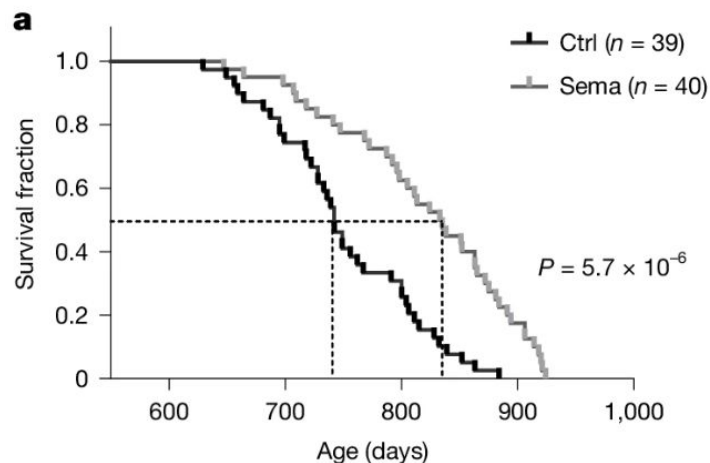
Multiple new GLP-1 medicines are under investigation for a wide range of indications.



Drucker, Daniel J. "The Expanding Landscape of GLP-1 Medicines." *Nature Medicine* 32, no. 1 (2026): 47–57. <https://doi.org/10.1038/s41591-025-04124-5>.

HbA1c: glycated haemoglobin A1c. Oral semaglutide Alzheimer's trials (EVOKE/EVOKE+) did not meet their primary endpoints (reported by the end of 2025).

Late-life semaglutide treatment slows ageing and extends lifespan in female mice



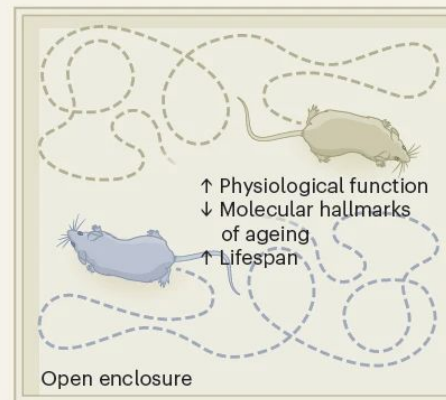
Feng *et al.* put 20-month-old female mice from an inbred laboratory strain on a calorie-restricted diet or treated them with semaglutide. The results suggest that GLP-1 receptor activation and calorie restriction slow ageing through overlapping but distinct mechanisms.

Y. Feng *et al.* (2026), "Late-life semaglutide treatment slows ageing and extends lifespan in female mice," <https://doi.org/10.1038/s41586-026-10940-7>, Nature; News and Views: [Weight-loss drug slows ageing in female mice](#) by Maria Emilia Fernandez & Rafael de Cabo

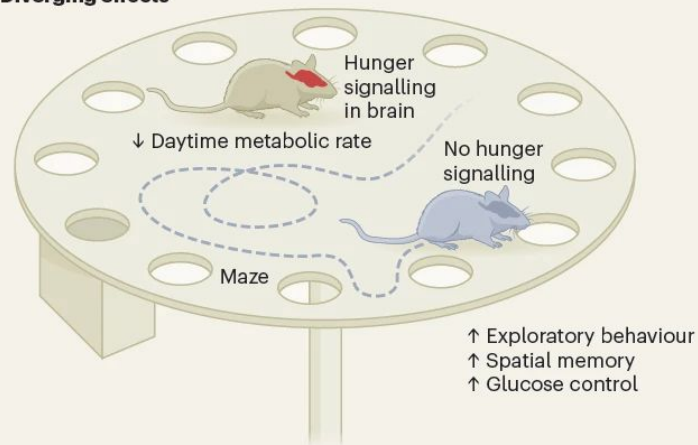
a Condition



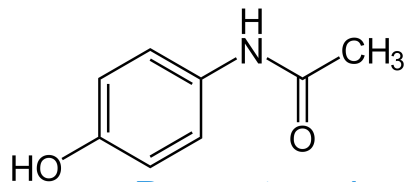
b Converging effects



c Diverging effects

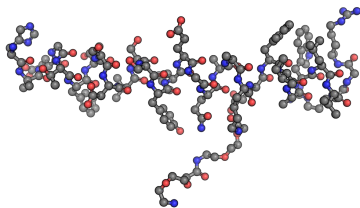


A zoo of modalities



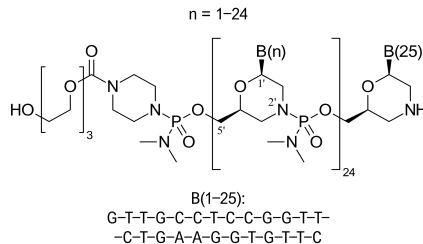
Paracetamol
Molecular weight
(MW): 151 Da

Small molecule



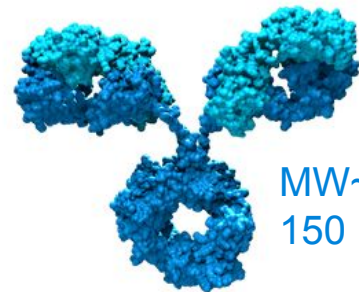
Semaglutide
MW~4 kDa

Peptide



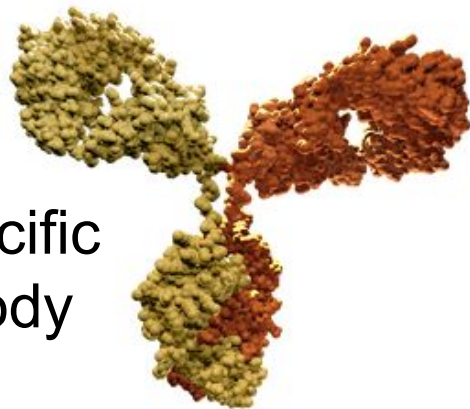
Golodirsen,
MW~5-30 kDa

Oligonucleotides

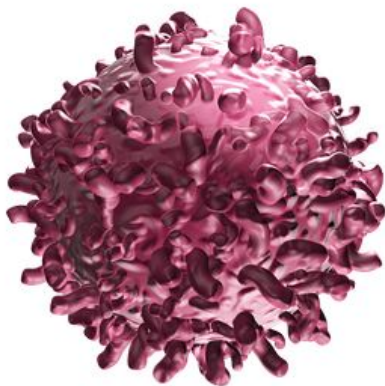


MW~
150 kDa

Monoclonal
antibody



Bispecific
antibody



Chimeric
Antigen
Receptor
(CAR) T
cells



mRNA vaccines

Novel drugs approved by the FDA's Center for Drug Evaluation and Research (CDER) in 2025

- Small molecules usually have a molecular weight (MW) less than 1000 Daltons.
- Oligonucleotides: MW between 5 and 30 kDa (5000-30000 Da), negatively charged
 - siRNA: small interfering RNA
- Proteins: MW ~150 kDa
 - mAb: monoclonal antibody
 - Bispecific: antibodies that bind simultaneously to two antigens or two epitopes of the same antigen.

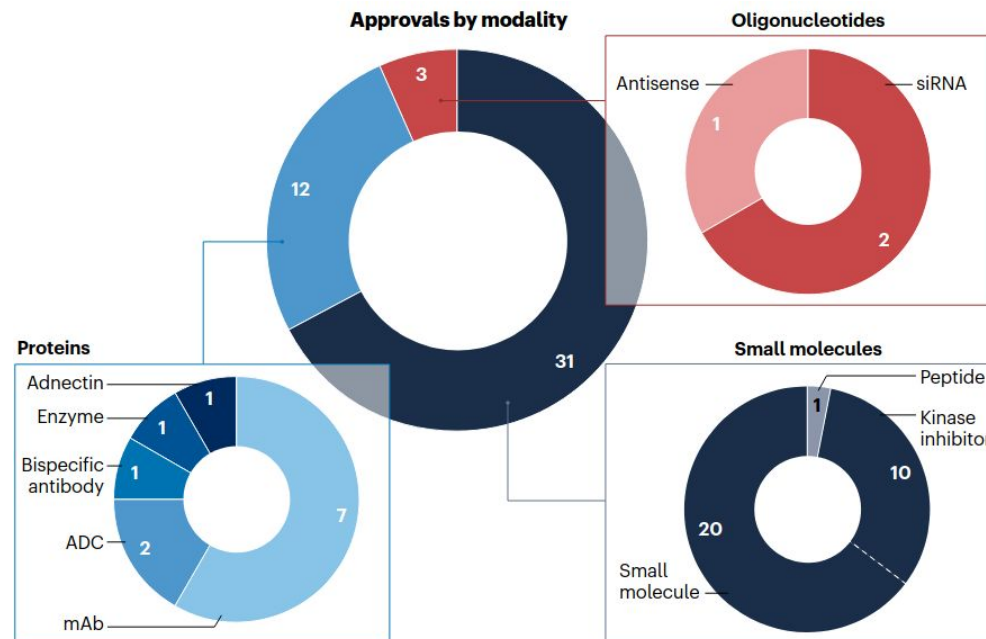
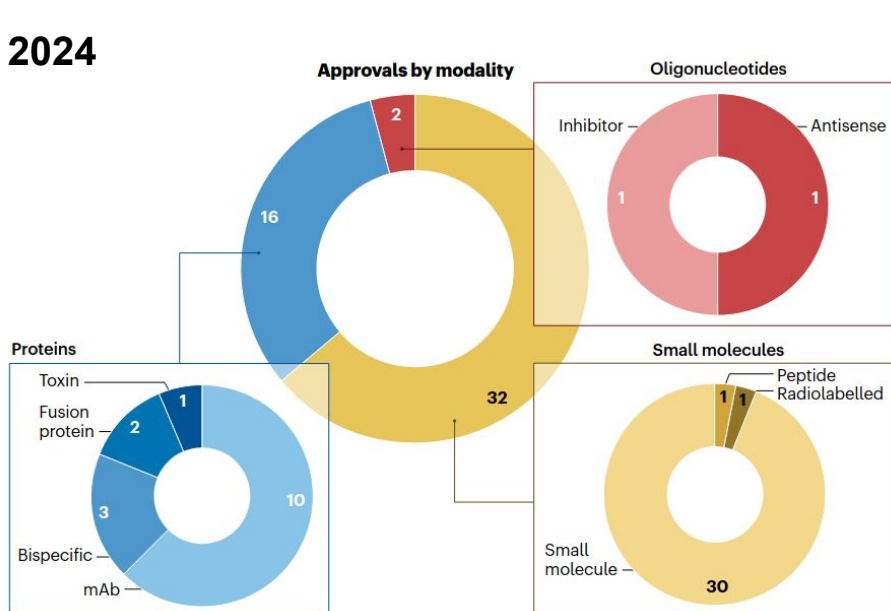


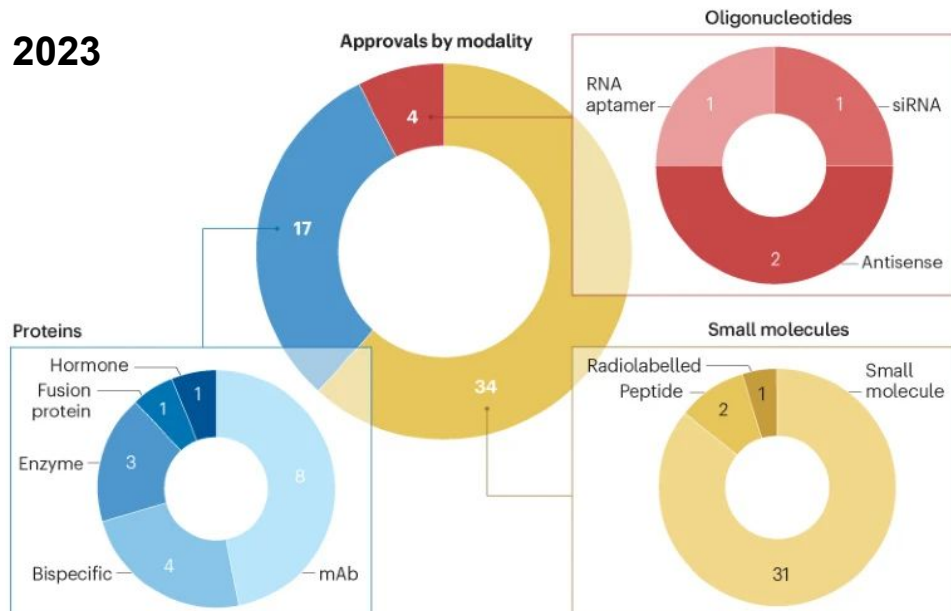
Fig. 3 | CDER approvals by modality. Small molecules, including peptides of up to 40 amino acids in length, and oligonucleotides are approved as new molecular entities. Protein-based candidates are approved through biologics license applications. ADC, antibody–drug conjugate; mAb, monoclonal antibody; siRNA, small interfering RNA. Source: *Nature Reviews Drug Discovery*.

Relative contributions of modalities have remained constant in the past three years

2024



2023



Approval data since 1993

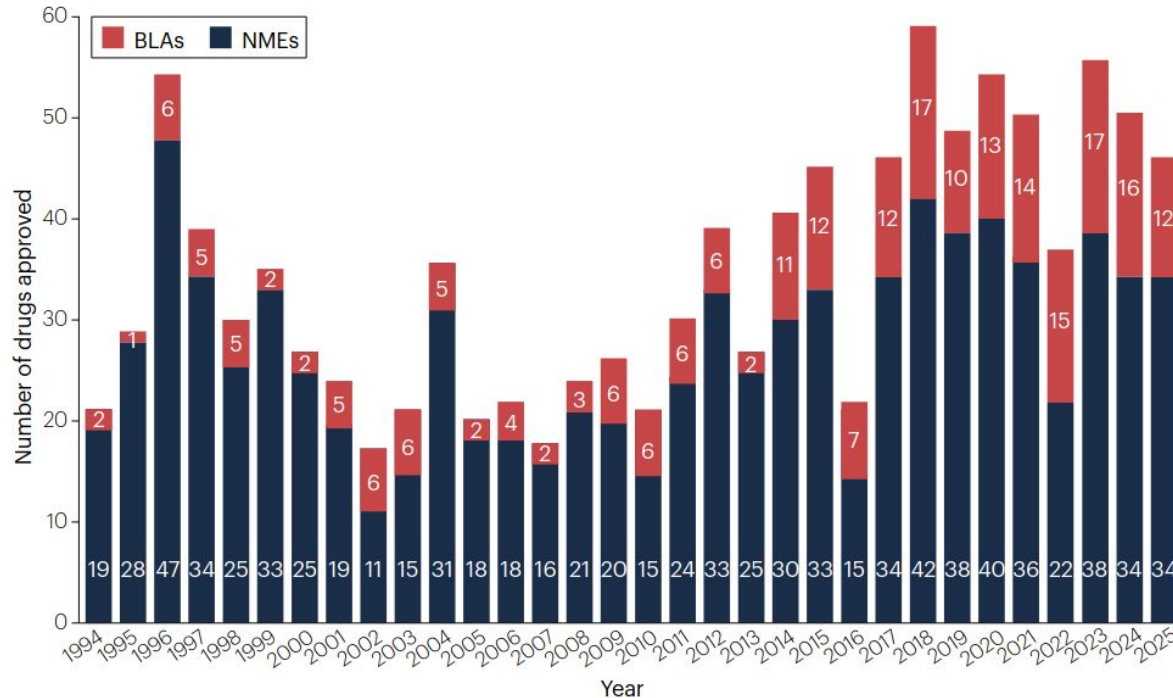


Fig. 1 | Novel FDA approvals since 1993. Annual numbers of new molecular entities (NMEs) and biologics license applications (BLAs) approved by the FDA's CDER. See Table 1 for new approvals in 2025. Products approved by CBER, including vaccines and gene therapies, are not included in this drug count (Table 2).

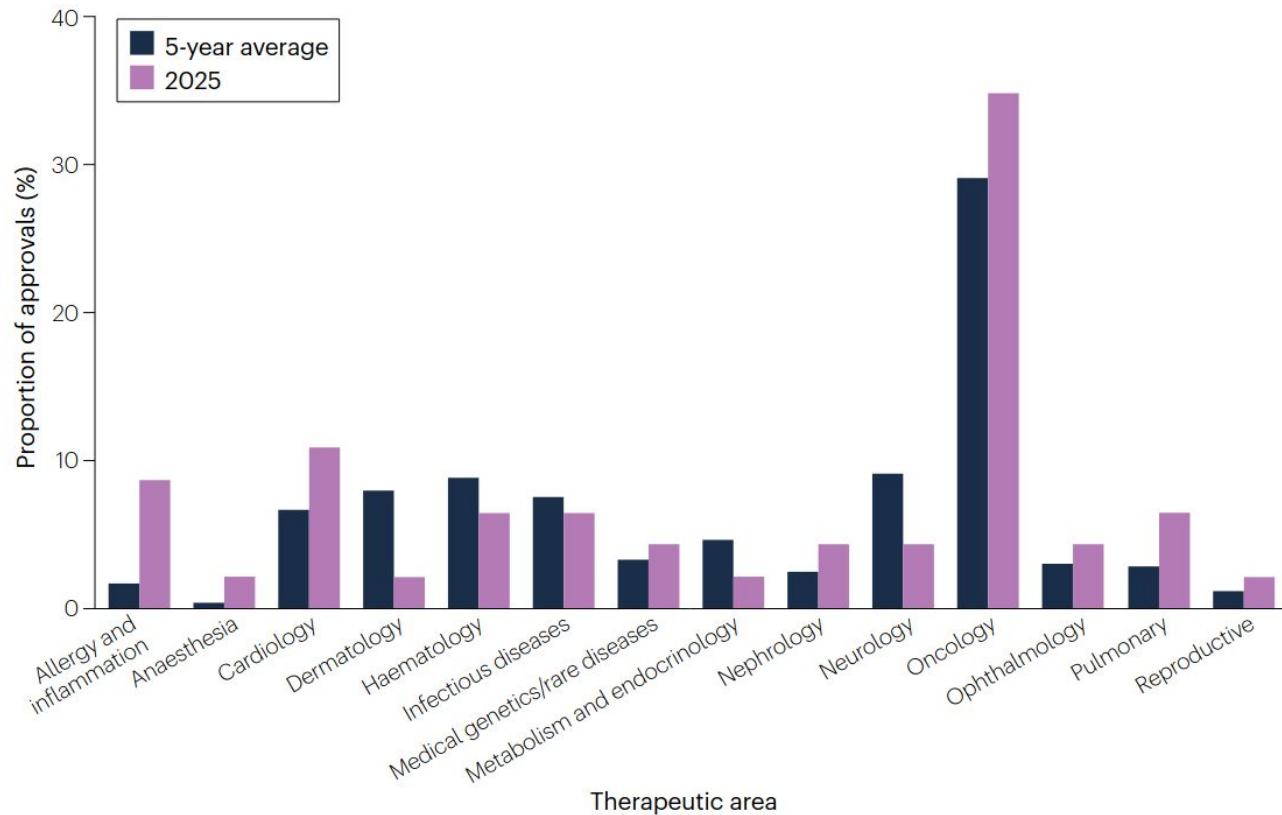
Source: FDA.

Approvals by FDA's Center for Drug Evaluation and Research (CDER):

- BLA: Biologics License Applications (mainly antibodies)
- NME: New Molecular Entities (small molecules, oligonucleotides).

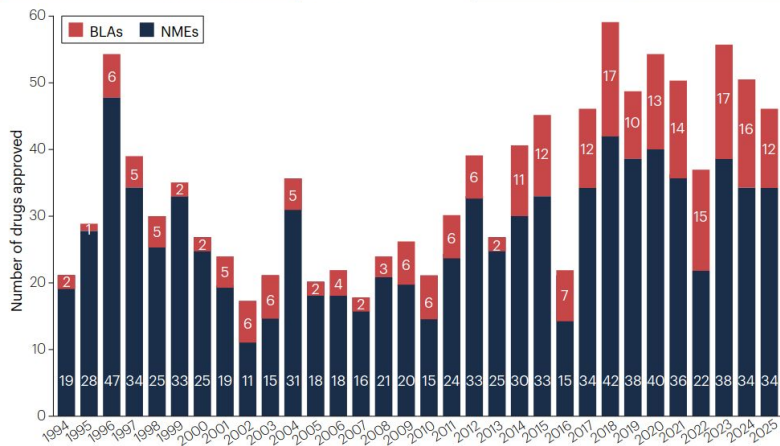
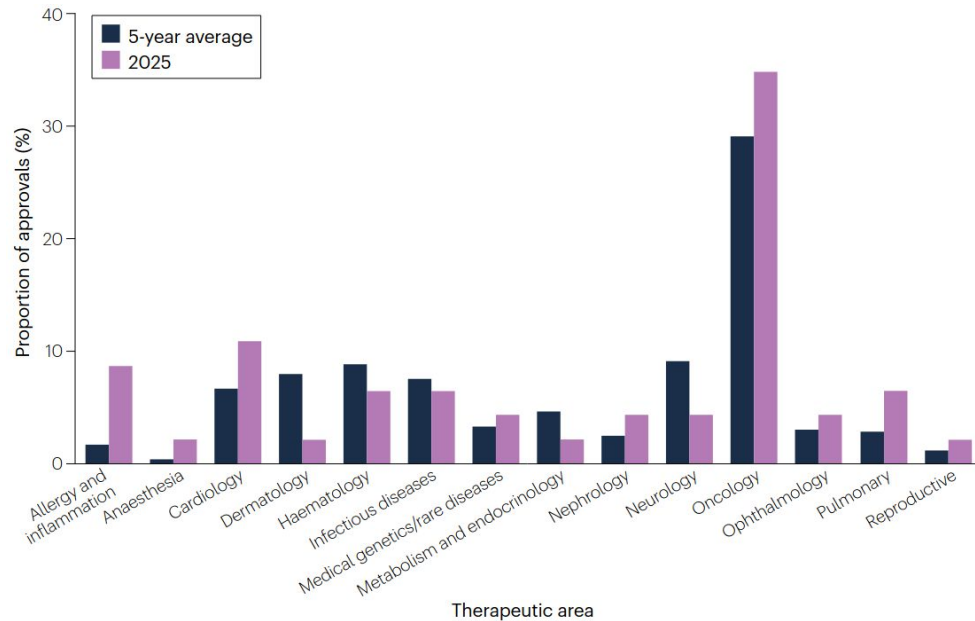
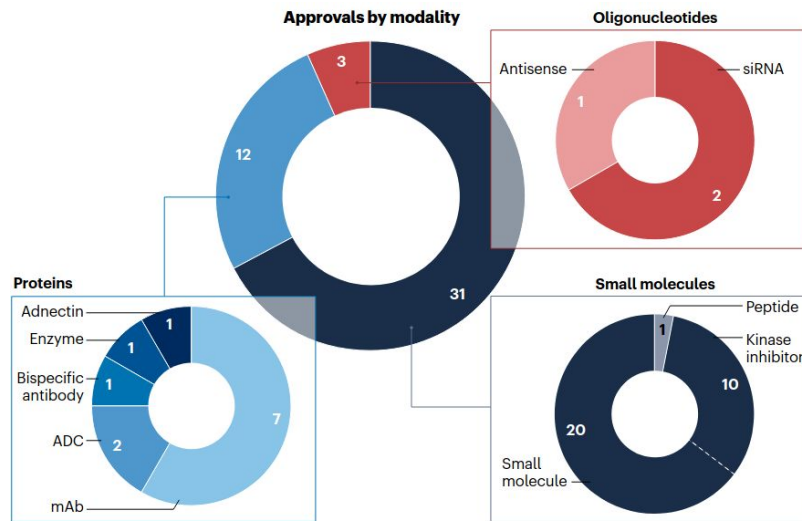
Vaccines and gene therapies are excluded.

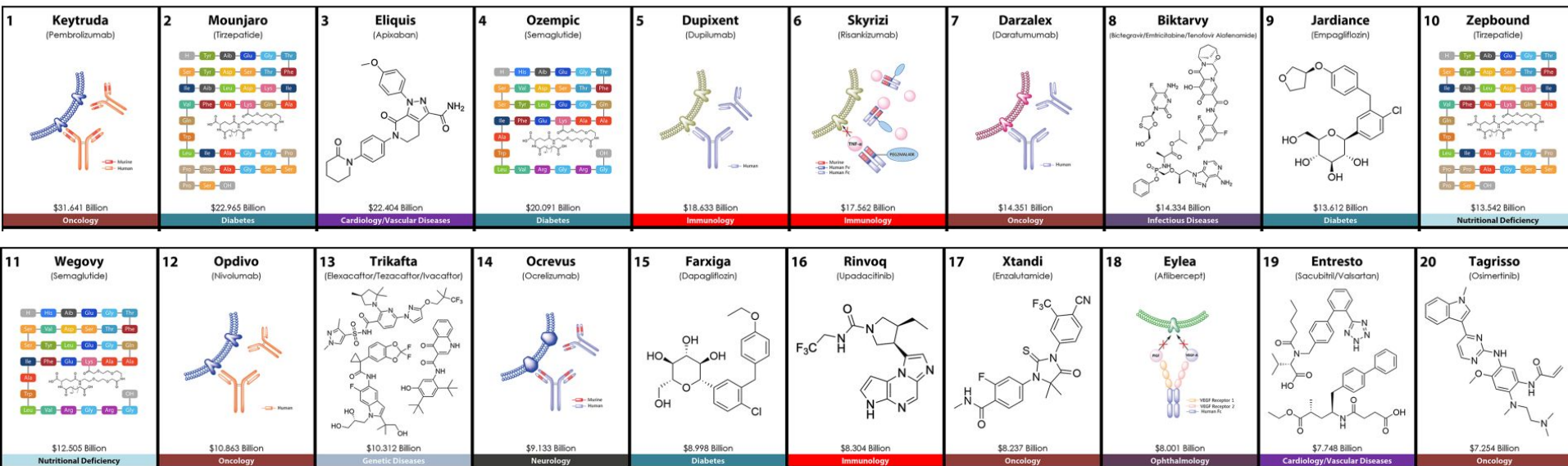
New drug approvals classified by therapeutic areas



- **Cardiology:** heart
- **Dermatology:** skin
- **Hepatology:** liver
- **Haematology:** blood
- **Infectious diseases:** infections due to viruses, bacteria, fungi, etc.
- **Medical genetics/rare diseases:** genetic disorders
- **Metabolism & endocrinology:** metabolism and hormones
- **Nephrology:** kidney
- **Neurology:** brain and peripheral nerves
- **Oncology:** cancer
- **Ophthalmology:** eye
- **Pulmonary:** lung
- **Reproductive:** reproduction and sex

What surprise you?





Top 20 pharmaceuticals by sales in 2025

Poster compiled by the Jon Njardarson group at University of Arizona (<https://njardarson.lab.arizona.edu>). Citation: J. Chem. Ed. 2010, 87, 1348.

Questions: (1) How many of them are small molecules, proteins, and oligonucleotides, respectively ? (2) Are there other modalities? (3) What patterns do you observe? (4) Do you have explanations for these patterns?

The social context of drug discovery: a role-playing game

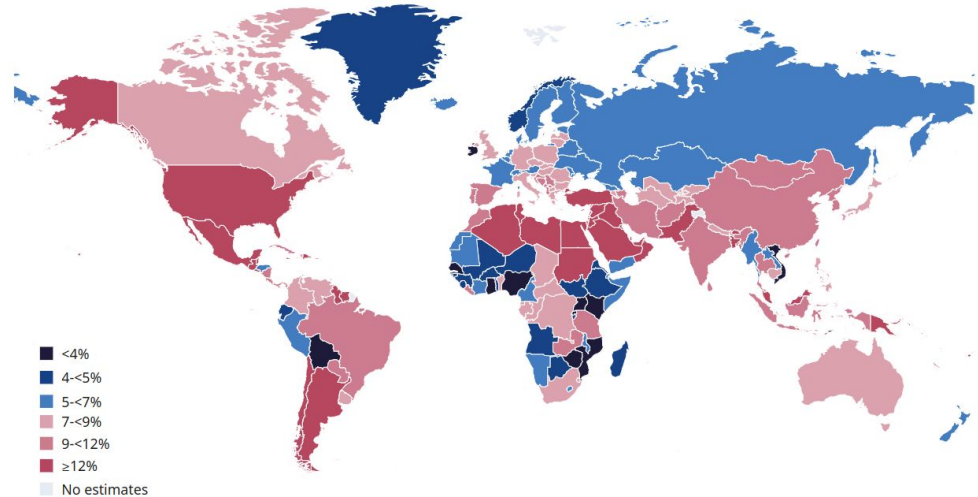
We consider a case study of developing new drugs to treat complications caused by type 2 diabetes, affecting ~10% of adults (20–79 y) worldwide (Switzerland: 4–6%, of whom ~28% are not diagnosed), with the possibility to expand it to the obesity indication.

We divide the classroom into the following personas:

1. Researchers (biology, medicine, AI, etc.): doing whatever research you want to do and getting funded
2. Research funding agency: giving fundings to any research problem, T2D is only one of them.
3. Patients: suffering, unfortunately, from T2D and its complications
4. Medical doctors: an medical expert in T2D who see patients regularly
5. Drug discovery company: determined to enter the T2D market
6. Insurance company: negotiating the price
7. The regulatory agency: drug approval

Questions: (1) What are your main interests and concerns? (2) With which groups do you wish to collaborate? Why? Rank the partners by priority. (3) What are the best-case and worst-case scenarios for you?

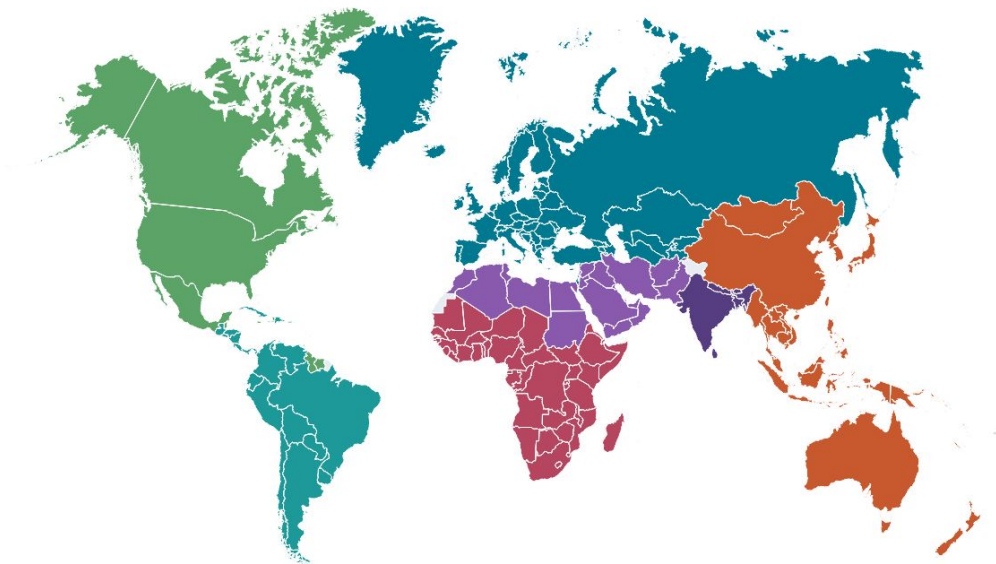
Map 3.2 Estimated age-standardised country prevalence of diabetes in adults (20–79 years), 2024.



[IDF Diabetes Atlas 2025](https://diabetesatlas.org/data-by-location/country/switzerland/) (11th version), The International Diabetes Federation (IDF). Swiss data is acquired from <https://diabetesatlas.org/data-by-location/country/switzerland/>

Statistics and forecast

Map 1 Number of people with diabetes worldwide and per IDF Region, in 2024–2050 (20–79 years)



World		
2050	852.5 Million	↑ 45% increase
2024	588.7 Million	

Europe (EUR)		
2050	72.4 Million	↑ 10% increase
2024	65.6 Million	

North America and Caribbean (NAC)		
2050	68.1 Million	↑ 21% increase
2024	56.2 Million	

South-East Asia (SEA)		
2050	184.5 Million	↑ 73% increase
2024	106.9 Million	

Africa (AFR)		
2050	59.5 Million	↑ 142% increase
2024	24.6 Million	

Middle-East and North Africa (MENA)		
2050	162.6 Million	↑ 92% increase
2024	84.7 Million	

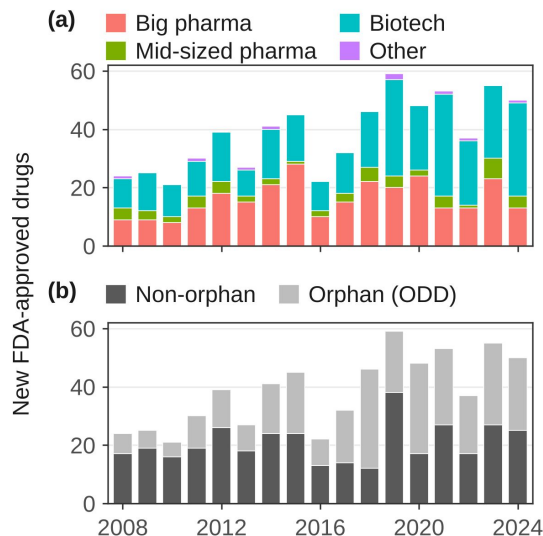
South and Central America (SACA)		
2050	51.5 Million	↑ 45% increase
2024	35.4 Million	

Western Pacific (WP)		
2050	253.8 Million	↑ 18% increase
2024	215.4 Million	

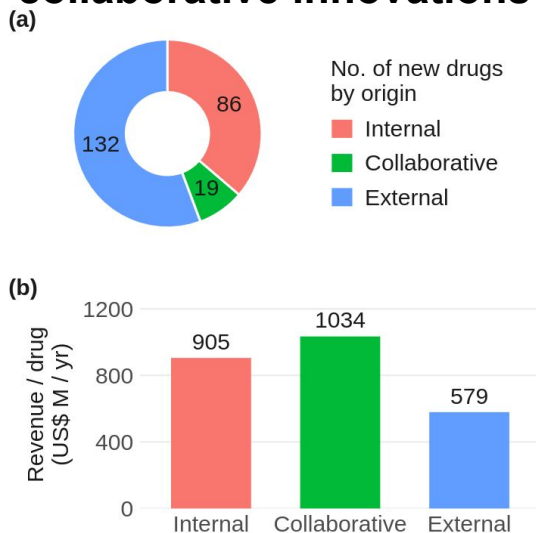
[IDF Diabetes Atlas 2025](#) (11th version)

Some changes beyond scientific progress in recent years

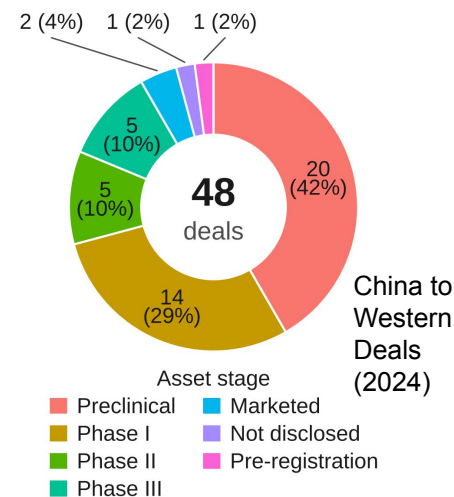
Rise of biotech and orphan indications



Pharma embracing external and collaborative innovations



Rise of R&D in China and emerging markets



Orphan indications: fewer than 1 patients out of 200,000 people (U.S. definition) Adapted from SSchuhmacher, Alexander, Oliver Gassmann, Sebastian Kwisda, Malte Kremer, Markus Hinder, and Dominik Hartl. "The R&D Productivity Challenge: Transforming the Pharmaceutical Ecosystem." *Drug Discovery Today* 30, no. 11 (2025): 104494. <https://doi.org/10.1016/j.drudis.2025.104494>; Schuhmacher, Alexander, Oliver Gassmann, Sara Knapp, Dominik Hartl, and Markus Hinder. "New Perspectives for Better R&D Performance: Lessons Learned from Leading Pharma Companies." *Drug Discovery Today* 31, no. 3 (2026): 104677. <https://doi.org/10.1016/j.drudis.2026.104677>; Tang, Leon 'Jun.' "Analysis of China-to-West Pharmaceutical Licensing Deals in 2024." *Nature Reviews Drug Discovery* 24, no. 6 (2025): 411–12. <https://doi.org/10.1038/d41573-025-00068-0>.